Multi-Disease Prediction Using Data Mining in Machine Learning.

Briti Dipan Srivastav, Gaurang Sharma, Jaydeep Davda
Department of Information Technology
Navrachana University
Vadodara, India

21/11/2022

I. Abstract

This research is based on prediction of the disease based on the symptoms of user facing and provides the accurate disease result based on that information. This research focuses in providing a better accuracy model which can detect multiple diseases that a person might suffering from. There are three algorithms used: decision tree, naive bayes and random forest. Among the above-mentioned algorithm naive bayes gives more accurate prediction and hence it can be taken for the final result. The dataset for the system is collected from the Kaggle website. The dataset has 132 features(symptoms) and 42 unique diseases. Hence the system can accurately predict 42 diseases. The accuracy given by the system is about 95% using naive bayes.

II. Introduction

Presently, when people suffer from a particular disease, then they have to visit a doctor which is very time consuming and costly too. Also, if the user is out of reach of doctors and hospitals it may be difficult for the user as the disease cannot be identified. So, if there is an automated program the process easier. The main goal of this research is to predict the disease beforehand so that the risk of the life can be prevented at an early stage and save life of people and the cost of treatment can reduce to a particular extent.

The healthcare industry can make effective decision making by "mining" the large database they possess i.e., by extracting the patterns and relationships hidden inside the database. According to [3] research there are 70% of people who suffer from general disease and 25% of people face death due to early ignorance. The main motive to develop the project is that the user can sit at their convenient place and have a check-up of their health.

The purpose of this research is to predict the accurate disease of the patient using all their general information and also the symptoms. Using this information, we will compare our previous datasets of the patient and predict the disease of the patient that they were suffering from. If this prediction is done at the early stages of the disease with the help of this project and all other necessary measures the disease can be cured and in general this predive system can also be very useful in the health industries.

The general purpose of this disease prediction is to provide prediction for the various and generally occurring disease that when unchecked and sometimes ignored can turn into fatal disease and can cause lot of problem to the patient as well as their family members.

III. Literature Survey

[1], in the present era the coronary heart disease which is similar type of heart disease has huge death rates. The dataset comprises of patients having future chance of CHD based on some feature. Many different algorithms are used with an aim to find which algorithm gives the best accuracy with each sampling concept.

[2], the main disease here is the skin disease which happens to the people of all the age groups for this application is developed to identify the type of skin disease. The main reason of happening a skin disease is conditions of immune system, having highly processed and ready meals and etc. The skin disease dataset contains different types of skin diseases which is divided into training and testing phases. The application made is a combination of disease detected and suggest medical treatment.

[3], Diabetes Mellitus DM is considered as the crucial disease due which many people suffer. The main factor for causing this disease is the unhealthy lifestyle, diet, heredity diabetes. In medical field to find a particular disease we have to know about its symptoms. The pipeline method is used for the model building.

[4], PCE use is consequently limited for patients with values outside the prespecified PCE ranges as well as patients without all variables available, preventing accurate guideline-based risk estimation to guide ASCVD risk reduction strategies. Since some EHR variables were not expected to have robust predictive power, a composite score incorporating the feature importance metrics returned by the tree-based methods as well as nonzero coefficients of LR Lasso was used to prune variables.

[5], in this paper, two more input attributes obesity and smoking are used to get more accurate result. Three datamining techniques were used and use of confusion matrix is done in order to calculate the accuracy of the model. The datamining tool Weka 3.6.6 is used for the experiment.

[6], the disease cerebrovascular was ranked the second death cause in Taiwan death every year. The research adopted different classification algorithms like decision tree, Bayesian classifier and back propagation neural network. The was to developed an optimal model for the disease with

analyzing and comparing classification efficiencies like sensitivity and accuracy.

[7], the disease taken into consideration was chronic kidney disease. The algorithms. The experimental comparison of the algorithms was done based on the performance measure of classification accuracy and execution time. Model testing and validation was performed by V-cross validation techniques. Missing predictor variable values were replaced by medians during the analysis.

[8], the disease focus is breast cancer. According to WHO many women suffer from the disease and also there are many death cases. The main aim to develop ML model to predict cancer at early stages. The confusion matrix is used to find the accuracy of the algorithms used.

[9], applies the information mining process predict high blood pressure from patient medical records. Under sampling techniques has been applied to come up with coaching knowledge sets, and data processing tool is used to improve the prediction performance.

[10], the main methodology used for this paper was through the survey of journals and publications in the field of medicine and computer science engineering. Tanagra tool is used to give an easy-to-use data mining software. As the result was very low, they used Weka 3.6.0 tool to enhance the prediction of classifiers.

[11], two different data mining techniques were used for the prediction of various disease and their performance was compared in order to evaluate the best classifier. Building precise and computationally efficient classifier is an important challenge in data mining and machine learning fields.

[12], Three main disease that were peak in USA in 2013 is heart attack, diabetes and breast cancer. Different datamining techniques were applied on the dataset to find the probabilistic value for a particular category mention above. The accuracy score was calculated with the help of the confusion matrix.

[13], a system that used the concept of datamining for generating a model that can easily predict heart disease. The use of MAFIA i.e., Maximal Frequent Itemset Algorithm is user for mining maximal frequent itemset from a transactional database.

[14], the research uses various techniques of datamining classifiers i.e., neural network, genetic and supervised algorithm. The Weka tool was used to enhance the accuracy in neural network and Tanagra tool is used in supervised learning algorithm.

[15], the research focus on predicting presence of heart disease with higher accuracy with a smaller number of attributes. The research involves the use of three classifiers and fuzzy logic is used for calculating the efficiency of the proposed system. Experiment was performing with Weka 3.6 tool.

[16], The relations of disease and source of disease & the consequences of signs are spontaneously measured in patients and assessed by users through DM approaches. The alike examination has recognized the disease of blood vessels and heart by statistical ethics and includes stroke, heart diseases, and peripheral artery disease, etc. This method uses the ID3 algorithm for training. Data reduction acquires condensed demonstration of database which is less in quantity and generates the same outcomes.

[17], chronic kidney disease is deadly and it is difficult to recognize at an early stage due to no symptoms. The main agenda of this work is to make a predictive model which will be able to predict weather the patient is suffering from chronic kidney disease or not.

[18], Here kidney disease data set is used and analyzed using Weka and Orange software. Al proposed performance evaluation of three data mining techniques for predicting kidney dialysis survivability. A data mining approach is used to extract knowledge about the interaction between the variables and patient survival. Feature space, extraction is taken place on rawdata.

[19], Diabetes Mellitus DM is considered as the crucial disease due which many people suffer. The main factor for causing this disease is the unhealthy lifestyle, diet, heredity diabetes. In medical field to find a particular disease we have to know about its symptoms. The pipeline method is used for the model building.

[20], It is better to identify a disease accurately on a proper time so the risk of a death of particular person disease. The dataset contains the disease and symptoms based of age and gender of a particular person. Machine helps to predict the disease more aaccurately and efficiently.

[21], at the present age the cardiovascular disease is growing day by day. So, to predict this kind of disease a request was made based on some feature like age, gender, cholesterol and blood pressure. Based on this different algorithm were used for the identification of this problem and kernel function was also used.

[22], the main disease taken into consideration is kidney which is also known as CDK and heart disease which is considered as a major problem in some humans. Different algorithms are used with the motive to find the initial stages of the disease which helps to reduce the risk of lives.

[23], Diseases are the main factor of unhealthy lifestyle many people in this world are diagnosed with different kind of diseases, so with the help of ML algorithms it is possible to cure them. Here the main focus is on the diseases like CKD, heart diseases, Breast cancer and lymphatic disease. For heart disease failure there was a system introduced to access the health of sick patients in this data mining was also used via cart classifiers.

[24], two classification models were applied on the dataset to predict the possibilities of having heart disease of a patient. The confusion matrix was used in order to calculate the accuracy generated.

[25], a computerized system checker is present to check and maintain health by knowing the symptoms. Also, there is a symptoms checker module that defines the body structure and gives liability to select the affected area and checkout the symptoms.

IV. Model Training

The model training is the process where we feed ML algorithm with data to help identify and learn good values for all attributes involved. The objective of training a machine learning model is to generate an accurate output or result based on some given attributes known as a target.

In model training the model tries to identify different patterns hidden in the dataset for the purpose of learning and uses this hidden pattern to identifying output to and unknown or new set of inputs.

We have taken three algorithms for training i.e., Decision tree, Random Forest and Native bayes classification algorithms because these algorithms are well known algorithms when come to solve classification problems and also our research is based on classification type problem.

During the model training we go through multiple stages like the following:

a) Data Preparation.

a. Identifying categorical data.

This is very important step as the machines cannot understand the textual data so it is important to identify the categorical data and handle it during the process. To know how much value each category have we have used value_counts() function as we have a category i.e., prognosis in the dataset we need to handle it.

```
dataset["prognosis"].value_counts()
```

```
Fungal infection                 120
Hepatitis C                      120
Hepatitis E                      120
Alcoholic hepatitis              120
Tuberculosis                     120
Common Cold                      120
Pneumonia                        120
Dimorphic hemmorhoids(piles)     120
Heart attack                     120
Varicose veins                   120
Hypothyroidism                   120
```

b. Handling of categorical data.

To handle the categorical data in the dataset we used the sklearn Label Encoder build in functions which allocated a series of numbers to each category and convert the text into number that can be recognized by the machine during the training.

```
#handling of categorical Data:
from sklearn.preprocessing import LabelEncoder
label = LabelEncoder()
dataset["prognosis"]=label.fit_transform(dataset["prognosis"])
dataset
```

c. Handling of missing values.

Missing value can bias the results of the machine learning models and reduce the accuracy of the model. There many different ways to handle the missing values in the dataset but for the research we have used sklearn Simple Imputer technique to handle missing data and replace is with the means value of the particular column.

```
#handling of missing data:

from sklearn.impute import SimpleImputer
imputer = SimpleImputer(missing_values=np.nan, strategy='mean')
imputer = imputer.fit(dataset)
dataset = imputer.transform(dataset)
```

d. Normalizing the dataset.

Normalization is a scaling technique in Machine Learning applied during data preparation to change the values of numeric columns in the dataset to use a common scale. It is not necessary for all datasets in a model. It is required only when features of machine learning models have different ranges.

```
#Normalizing the dataset:
for cols in dataset.columns:
    for datatype in dataset.dtypes:
        if datatype =='int64':
            dataset[cols] = dataset[cols]/dataset[cols].max()
```

b) Splitting data into training and testing data.

It is the process of dividing the data into training data that can be used to train the data and testing data that can be used to train the data. To do so we have used the sklearn train_test_split library used to split the data. The dataset is spilt on random basis between the training and testing data. By default, the Test set is split into 30 % of actual data and the training set is split into 70% of the actual data.

```
#Spliting data into training and testing.
from sklearn.model_selection import train_test_split
x = dataset.iloc[:,:-1].values
y = dataset.iloc[:,-1:].values

xtrain, xtest, ytrain, ytest = train_test_split(x, y, test_size=0.3, random_state = 0)
```

c) Training data using Decision Tree Algorithm.

We have used sklearn in order to train the model with decision tress algorithms which uses the train and test data to generate outputs. To do so following code is used.

```
#Model training using Decision Tree algorithm:

from sklearn.tree import DecisionTreeClassifier
dc = DecisionTreeClassifier(max_depth =5, random_state = 0)
dc.fit(xtrain, ytrain)

ypred= dc.predict(xtest)

ycal = pd.DataFrame({'Actual':ytest.flatten(),'Predict':ypred.flatten()})
print(ycal)
```

d) Training data using Random Forest Algorithm.

Again, we have used sklearn to train the model with random forest algorithm which uses the train and test to generate outputs. To do so following code is used.

```
#Model training using Random Forest algorithm:

from sklearn.ensemble import RandomForestRegressor
regressor = RandomForestRegressor(n_estimators = 100, random_state = 0)

regressor.fit(xtrain, ytrain)
y_pred =regressor.predict(xtrain, ytrain)


ycal = pd.DataFrame({'Actual':ytest.flatten(),'Predict':ypred.flatten()})
print(ycal)
```

e) Training data using Native Bayes Algorithm.

Again, we have used sklearn to train the model with native bayes algorithm which use the train and test to generate outputs. To do so following code is used.

```
#Model training using Native Bayes algorithm:

from sklearn.naive_bayes import GaussianNB
gnb = GaussianNB()

gnb.fit(xtrain, ytrain)
y_pred = gnb.fit(xtrain, ytrain).predict(xtest)


ycal = pd.DataFrame({'Actual':ytest.flatten(),'Predict':ypred.flatten()})
print(ycal)
```

Once the model training is finalized with a proper dataset along with the algorithm used in the research that can be used for result analysis and father processing.

V. Result Analysis

The result analysis is the phase where we find the best accuracy between the different algorithms taken into consideration and try to finalize the conclusions based on the result generated.

This is the final stage of any research were based on every analysis and practical implementation we extract different result and finally based on the result we come up with certain conclusions.

In the research we have use the confusion matrix to calculate the accuracy or result that we want and the accuracy generated by each algorithm are follow:

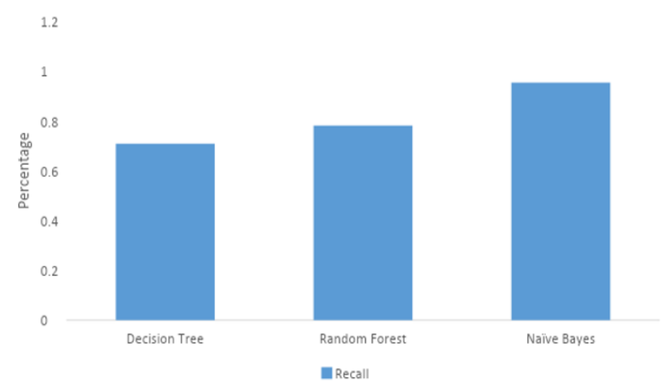
A. *Recall Comparision of Algorithms:*

The recall is calculated as the ratio between the numbers of Positive samples correctly classified as Positive to the total number of Positive samples. The recall measures the model's ability to detect positive samples. The higher the recall, the more positive samples detected.

We have used metrics library in sklearn to construct a confusion matrix and the formula used to calculate recall is as follow:

Recall = True Positive/True Positive + False Negative

Recall = TP/TP+FN

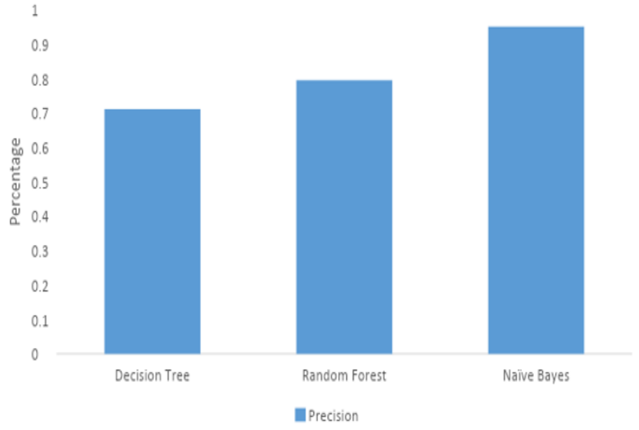


B. *Precision comparision of Algorithms:*

Precision is how good the model is at predicting a specific category.

We have used metrics library in sklearn to construct a confusion matrix and the formula used to calculate precision is as follow:

precision= True Positive/True Positive + False Positive
precision=TP/TP+FP



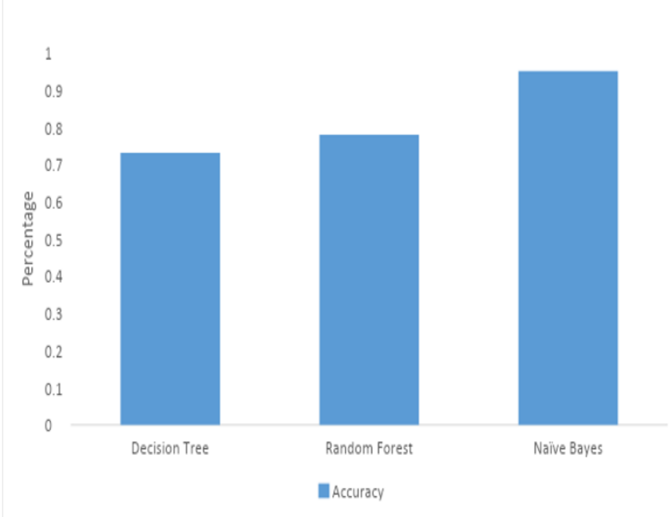
C. *Accuracy Comparision of Algorithms:*

Accuracy is one metric for evaluating classification models. Informally, accuracy is the fraction of predictions our model got right. Accuracy can be calculated in terms of positives and negatives as follows:

Accuracy = TP + TN / TP + TN + FP + FN

Where TP = True Positives, TN = True Negatives, FP = False Positives, and FN = False Negatives.

The Accuracy finally tell how good the machine learning model performs on the given data when an unknown or new data arrived to the system.



	Accuracy	Precision	Recall
Decision Tree	0.73	0.71	0.70
Random Forest	0.78	0.79	0.78
Naive Bayes	0.95	0.95	0.95

Table 1: The Above table shows the accuracy, precision and recall generated by corresponding algorithms. The use of confusion matrix also helps to understand the model working more in details. Here we can easily analyze the different accuracy generated by Decision tree, Random Forest and Native bayes algorithms and can easily conclude that Native Bayes algorithm produce the best result in the research analysis.

VI. CONCLUSION

The research was aim to create a machine learning algorithm that will be best for predicting multi disease that were supposed to be taken into consideration unless it become fatal or dangerous to health. We have used three classification algorithms i.e., Decision tree, Random Forest and Native bayes which are well knows algorithms for classification types problems that we need as per our requirements. We have used different sklearn libraries to train different algorithms and also to calculate different result like Recall, precision and Accuracy. After doing lots of analysis and practical implementation we are able to finally conclude that out of all the algorithm taken into consideration Native Bayes algorithm performs best with highest accuracy.

VII. REFERENCES

[1] A. Lakshmanarao, Y. Swathi, P. Sri Sai Sundareswar., Machine Learning Techniques for Heart Disease Prediction, 2019.

[2] Ahmed A. Elngar, Rishabh Kumar, Amber Hayat, Prathamesh Churi., Intelligent System for Skin Disease Prediction using Machine Learning., 2021.

[3] Aishwarya Majumdar, Dr. Vaidehi., Diabetes Prediction using Machine Learning Algorithms., 2020.

[4] Andrew Ward, Ashish Sarraju, Sukyung Chung, Jiang Li, Robert Harrington, Paul Heidenreich,, Machine learning and atherosclerotic cardiovascular disease risk prediction in a multi-ethnic population., 2020.

[5] Chaitrali S dangare, Sulabha S Apte., Improved Study of Heart Disease Prediction System using Data Mining Classification Techniques., 2012.

[6] Duen-Yian Yeh, Ching-Hsue Cheng., A predictive model for cerebrovascular disease using data mining., 2011.

[7] El – Houssainy, A.Rady. Ayman S. Anwar., Prediction of kidney disease stages using data mining algorithms., 2019.

[8] F. M. Javed Mehedi Shamrat, Md. Abu Raihan, A.K.M. Sazzadur Rahman, Imran Mahmud, Rozina, An Analysis on Breast Diseases prediction using Machine Learning Approaches., 2020.

[9] Feixiang Huang, S. W.-C., Predicting Disease by Using Data Mining Based on Healthcare Information System., 2012.

[10] Jyoti Soni, Ujma Ansari, Dipesh Sharma, Sunita Soni., Predictive Data Mining for Medical Diagnosis: An Overview of Heart Disease Prediction., 2011.

[11] K. Gomathi, D. S., Multi Disease Prediction Using Data Mining Techniques., 2016.

[12] K.Ashok, M. Mounika, S.Rasheeduddin., Multi Disease Prediction using Data Mining Techniques., 2012.

[13] M.A. Nishara Banu, B. Gomathy., DISEASE PREDICTING SYSTEM USING DATA MINING TECHNIQUES., 2013.

[14] Nidhi Bhatia, K. J. , A Novel Approach for Heart Disease Diagnosis using Data Mining and Fuzzy Logic., 2012.

[15] Nidhi Bhatia, Kiran Jyoti., An Analysis of Heart Disease Prediction using Different Data Mining Techniques., 2012.

[16] P Tiwari, V Singh., Diabetes disease prediction using significant attribute selection and classification approach., 2020.

[17] Pal, Saurabh., Chronic Kidney Disease Prediction Using Machine Learning Techniques., 2022.

[18] Parul Sinha, Poonam Sinha., Comparative Study of Chronic Kidney Disease Prediction using KNN and SVM., 2015.

[19] Ravinder Ahuja, Subhash C Sharma, Maaruf Ali., A Diabetic Disease Prediction Model Based on Classification Algorithms., 2020.

[20] Rinkal Keniya, Aman Khakharia, Vruddhi Shah, Vrushabh Gada, Ruchi Manjalkar, Tirth Thaker,Mahesh Warang, Ninad Mehendale., Disease Prediction from various symptoms using machine learning., 2020.

[21] Rubini PE1, Dr.C.A.Subasini, Dr.A.Vanitha Katharine, V.Kumaresan, S.GowdhamKumar, T.M., A Cardiovascular Disease Prediction using Machine Learning Algorithms., 2021.

[22] Salkar, Claire., A Detailed Analysis on Kidney and Heart Disease Prediction using Machine Learning., 2021.

[23] Shubham Gupta, Vishal Bharti, Anil Kumar., A Survey on various Machine Learning, 2019.

[24] Sonam Nikhbar, A.M. Karandikar., Prediction of Heart Disease Using Machine Learning Algorithms., 2016.

[25] Tomar, A., An approach to devise an Interactive software solution for smart health prediction using data mining., 2016.